

Ex. 14. The average of five numbers is 58. The average of the first two numbers is 48.5 and the average of last two numbers is 53.5. What is the third number? (Bank P.O., 2009)

Sol. Third number = (Sum of 5 numbers) – (Sum of 4 numbers)
 $= (58 \times 5) - [(48.5 \times 2) + (53.5 \times 2)]$
 $= 290 - (97 + 107) = 290 - 204 = 86.$

Ex. 15. The average of marks of 13 papers is 40. The average of marks of the first 7 papers is 42 and that of the last 7 papers is 35. What are the marks of the seventh paper? (C.P.F., 2007)

Sol. Marks in the seventh paper = $(42 \times 7 + 35 \times 7 - 40 \times 13)$
 $= (294 + 245 - 520) = 539 - 520 = 19.$

Ex. 16. The average age of A and B is 20 years. If A is replaced by C, the average age becomes 19 years and if B is replaced by C, the average age becomes 21 years. Find the ages of A, B and C. (S.S.C., 2006)

Sol. Clearly, average age of A and B = 20 years;

average age of B and C = 19 years;

average age of A and C = 21 years.

Sum of ages of A and B = (20×2) years = 40 years ...(i)

Sum of ages of B and C = (19×2) years = 38 years ...(ii)

Sum of ages of A and C = (21×2) years = 42 years ...(iii)

Adding (i), (ii) and (iii), we get:

2 (Sum of ages of A, B and C) = 120 years

$\Rightarrow$ (Sum of ages of A, B and C) = 60 years ...(iv)

Subtracting (i) from (iv), we get: C's age = $(60 - 40)$ yrs = 20 years

Subtracting (ii) from (iv), we get: A's age = $(60 - 38)$ yrs = 22 years

Subtracting (iii) from (iv), we get: B's age = $(60 - 42)$ yrs = 18 years

Ex. 17. In a class there are 50 students. Their average weight is 45 kg. When a student leaves the class, the average is reduced by 100 g. Find the weight of the student who left the class. (R.R.B., 2006)

Sol. Total weight of 50 students = (45×50) kg = 2250 kg.

Average weight of 49 students = $(45 \text{ kg} - 100 \text{ g}) = 44.9 \text{ kg}.$

Total weight of 49 students = (44.9×49) kg = 2200.1 kg.

$\therefore$ Weight of the student who left the class = $(2250 - 2200.1)$ kg = 49.9 kg.

Ex. 18. The average age of a class of 39 students is 15 years. If the age of the teacher be included, then the average increases by 3 months. Find the age of the teacher.

Sol. Total age of 39 persons = (39×15) years = 585 years.

Average age of 40 persons = 15 years 3 months = $\frac{61}{4}$ years.

Total age of 40 persons = $\left(\frac{61}{4} \times 40\right)$ years = 610 years.

$\therefore$ Age of the teacher = $(610 - 585)$ years = 25 years.

Ex. 19. The average of 11 players of a cricket team is increased by 2 months when two of them aged 18 years and 20 years are replaced by two new players. Find the average age of the new players. (S.S.C., 2007)

Sol. Total age increased = (2×11) months = 22 months = 1 years 10 months.

Sum of ages of two new players = $(18 \text{ years} + 20 \text{ years} + 1 \text{ years } 10 \text{ months})$
 $= 39 \text{ yrs } 10 \text{ months}.$

$\therefore$ Required average = $\left(\frac{39 \text{ years } 10 \text{ months}}{2}\right) = 19 \text{ years } 11 \text{ months}.$

Ex. 20. The average age of a class was 15 years. When 5 boys whose average age was 12 years 6 months were admitted in the class, the average was reduced by 6 months. How many students were there in the class originally? (P.C.S., 2008)

Sol. Let the original number of students in the class be x . Then,

$$15x + 12\frac{1}{2} \times 5 = 14\frac{1}{2}(x + 5) \Rightarrow 15x + \frac{25}{2} \times 5 = \frac{29}{2}x + \frac{29}{2} \times 5 \Rightarrow \frac{x}{2} = 10 \Rightarrow x = 20.$$

Hence, required number of students = 20.